

Project Initialization & Planning Phase: Proposed Solution

Comprehensive proposal for ComicCraft – AI Comic Story Creator using Gemini Models

Project Name:	ComicCraft - AI Comic Story Creator using Gemini Models	Date:	30 September 2026
Team ID:	SWTID-2026-5650	Maximum Marks:	5 Marks

1. Proposal Summary

The proposal report aims to transform comic story creation using Google Gemini models, boosting authoring efficiency, visual story consistency, and audience engagement. It tackles traditional illustration bottlenecks, promising automated script-to-panel rendering, context-aware character persistence, and interactive digital publishing. Key features include Gemini multimodal script parsing, sequential panel layout generation, and real-time dialogue bubble synthesis.

2. Project Overview & Problem Statement

Section	Details & Scope
Objective	Build ComicCraft, an end-to-end AI comic creation platform using Gemini models to automate script parsing, panel layout, visual art prompting, and dialogue placement.
Scope	Comprehensively covers story idea expansion, multimodal script breakdown, character visual memory persistence across panels, speech bubble overlay, and digital comic PDF/web export.
Problem Description	Traditional comic production requires expert illustration skills, extensive drawing time (10–20 hrs/page), and high art budgets, creating severe entry barriers for non-artist storytellers.
Impact	Solving these issues democratizes visual storytelling, reduces comic production timelines by 90%, lowers costs, and enables rapid iteration of indie webtoons and educational graphic novels.
Proposed Approach	Employing Google Gemini 1.5 Pro / Flash models for multimodal script understanding, character consistency tracking, automated visual prompt generation, and sequential panel layout synthesis.

Key Features of Proposed Solution:

- Gemini Multimodal Script & Dialogue Parser.
- Sequential Character Consistency & Style Anchor Engine.
- Dynamic Multi-Panel Layout Generator with Auto-Speech Bubbles.
- Real-time Interactive Creator Canvas & Web/PDF Viewer.
- Continuous learning and story template adaptation.

3. Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	NVIDIA T4 / A10G GPU (Cloud/Local), 8 vCPU Cores

Hardware	Memory (RAM)	16 GB System RAM, 16 GB VRAM
Hardware	Storage Disk	500 GB High-Speed NVMe SSD
Software	Frameworks	FastAPI, Streamlit / React, PyTorch, LangChain
Software	Libraries	google-generativeai SDK, Pillow, OpenCV, ReportLab, Pandas, NumPy
Environment	IDE / Dev Tools	VS Code, Jupyter Notebook, Google Colab
Data	Datasets & Prompts	Custom Comic Script & Panel Datasets, Gemini Multimodal Prompts (10,000+ panels, JSON/CSV)